Maximum likelihood and MAP: why squared error and cross-entropy follow from probability assumptions

When a language model is trained, what exactly is it trying to make as large as possible?

Mathematics for AI and Machine Learning | Session 16 of 18 | Probability and statistics
KLAi Professional Learning | Jotham Martin Wambi

Today in one view: what you will leave with

By the end of today you can



Find the parameter that makes the data most likely (maximum likelihood estimation, MLE) for a coin and for a Gaussian mean, by hand and by scanning

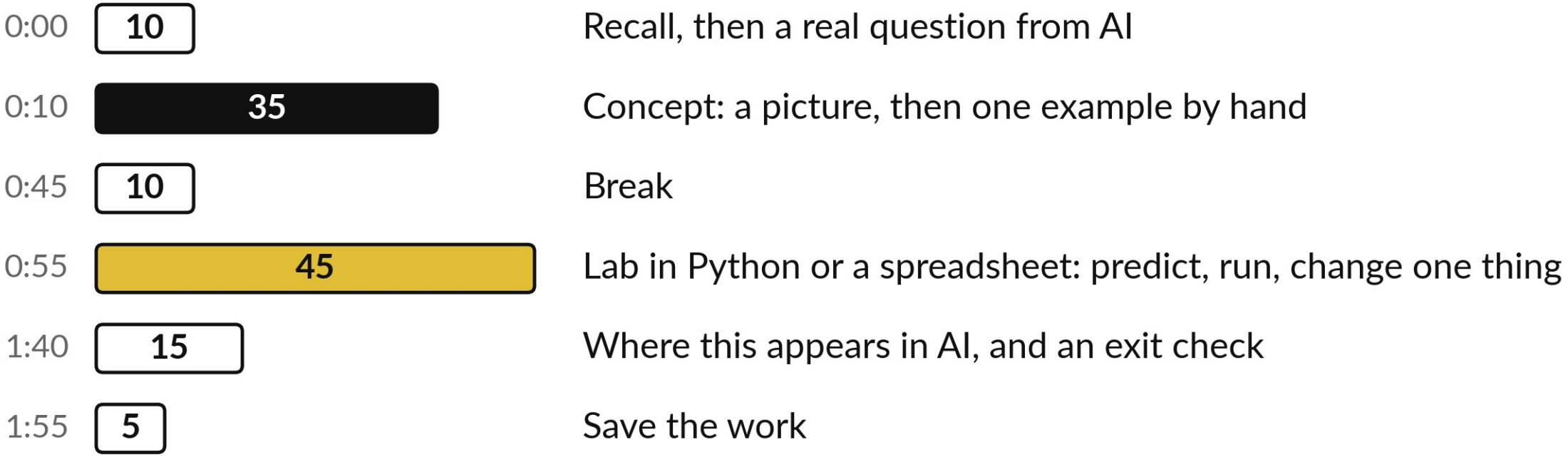


Show that squared error comes from Gaussian noise (with the same sigma for every row) and cross-entropy from a Bernoulli or categorical assumption



Explain MAP as likelihood times a prior, its link to weight decay, and the trade-off between bias and variance

Your two hours: likelihood, loss functions, then AI



Ideas first. Then the lab, in Python or a spreadsheet. Then AI.

Three questions from last time

1. What does P(A | B) mean, in words?

2. State Bayes' theorem in words.

3. Why can a positive result from an accurate test still mean a small chance of disease?



Answer aloud first. Then we check together.

Answers to the three questions from Session 15

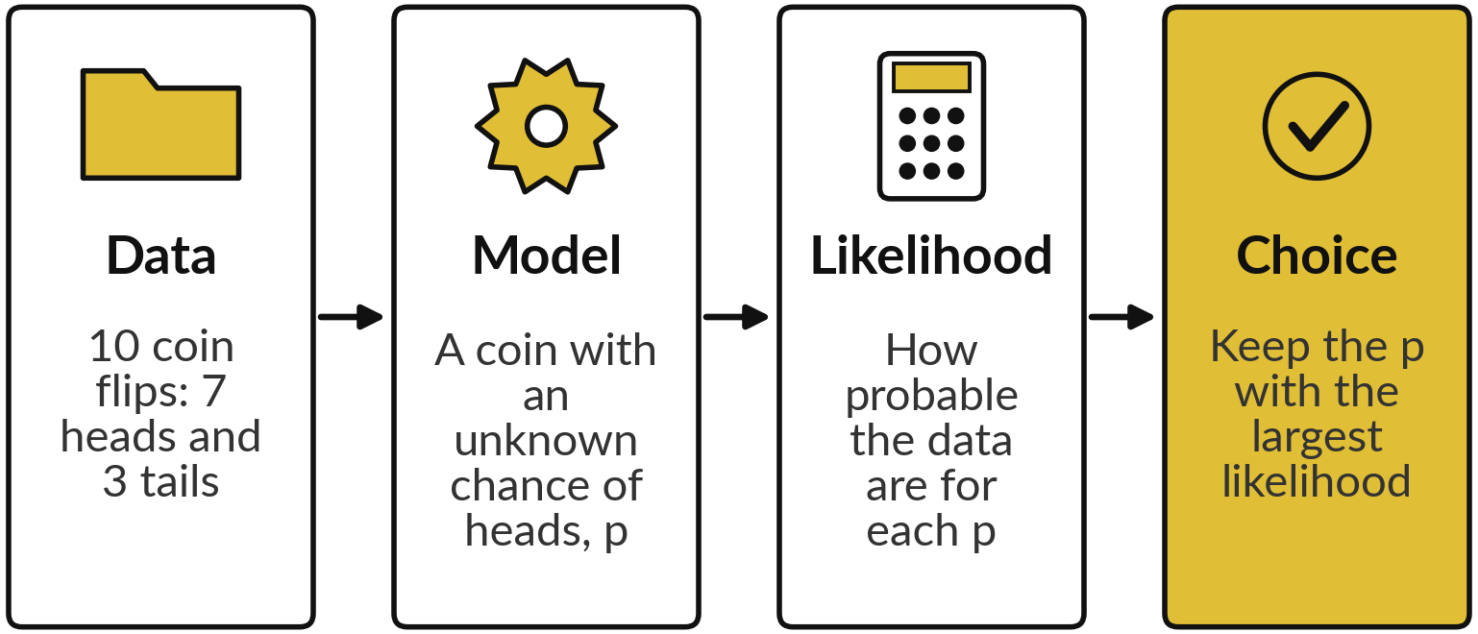
Question	Short answer
What does P(A | B) mean, in words?	The probability of A given that B has happened. Look only at the cases where B is true, and take the fraction of them in which A is also true.
State Bayes' theorem in words.	The posterior equals the likelihood times the prior, divided by the evidence: P(A | B) = P(B | A) × P(A) / P(B).
Why can a positive result from an accurate test still mean a small chance of disease?	Because the disease is rare. The prior is small, so most positive results are false alarms. The prior matters.

Today we use the prior and the likelihood from the third answer.

Words we use today

	Word	What it means
	**Likelihood**	How probable the exact data we saw are, for one chosen value of the parameter. Written L(theta).
	**Maximum likelihood estimation (MLE)**	Choose the parameter value with the largest likelihood.
	**Log-likelihood**	The natural logarithm of the likelihood. A product becomes a sum.
	**Prior and MAP**	The prior is what we believe before the data. Maximum a posteriori (MAP) estimation maximises likelihood times prior.
	**Bias and variance**	Bias: the error of the average prediction. Variance: how much the prediction moves between training sets.

Training means choosing the parameter values that make the data we saw most probable

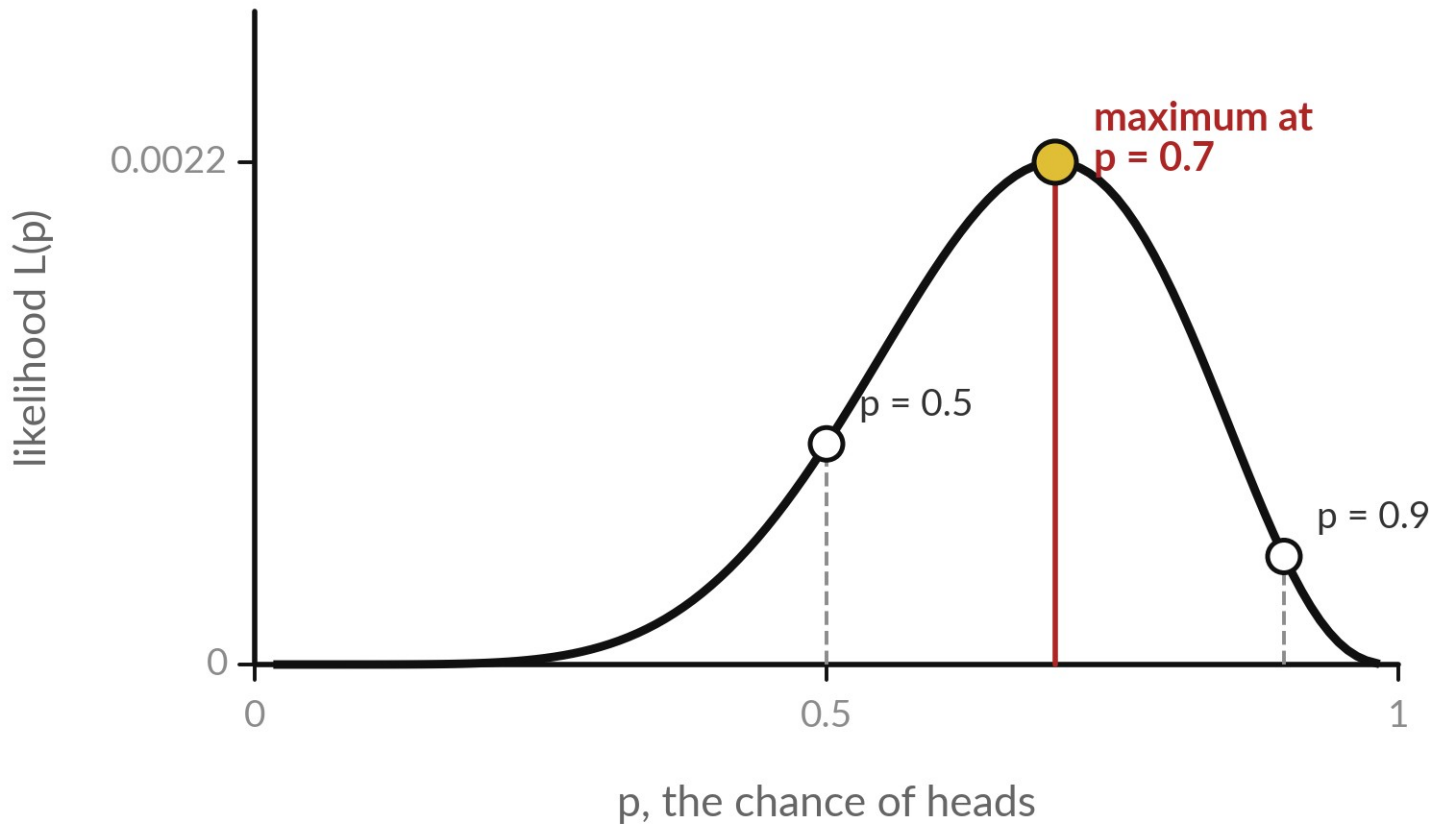


Maximum likelihood estimation (MLE) picks the parameter that makes the observed data most probable.

A parameter is a number the model learns, such as p, the chance of heads.

Today the model is a coin, a Gaussian curve and a classifier.

The likelihood fixes the data and varies the parameter



The likelihood L(p) is how probable our exact sequence of 10 flips (7 heads, 3 tails, in the order we saw) is if the chance of heads is p.

Each point on the curve is one candidate p. The height is the likelihood of the same data.

The curve peaks at p = 0.7, which is 7 heads out of 10.

By hand: the likelihood of one sequence with 7 heads and 3 tails, for three values of p

Flips are independent, so the likelihood of this exact sequence is a product: p for each head, 1 - p for each tail.

p = 0.5	0.5^7 × 0.5^3 = 0.000977
p = 0.7	0.7^7 × 0.3^3 = 0.0823543 × 0.027 = 0.002224
p = 0.9	0.9^7 × 0.1^3 = 0.4782969 × 0.001 = 0.000478
Any order	For 7 heads in any order, multiply by 120 (10 choose 7): 120 × 0.002224 = 0.2668, the Session 14 value. The peak stays at 0.7.

Scan a grid of p values: the likelihood of our exact sequence and its log are both largest at p = 0.7

p	Likelihood L(p)	Log-likelihood ln L(p)	p	Likelihood L(p)	Log-likelihood ln L(p)
0.1	0.000000	-16.434	**0.6**	0.001792	-6.325
0.2	0.000007	-11.935	**0.7**	0.002224	-6.109
0.3	0.000075	-9.498	**0.8**	0.001678	-6.390
0.4	0.000354	-7.947	**0.9**	0.000478	-7.645
0.5	0.000977	-6.931			

For a coin the MLE has a closed form: the fraction of heads

Likelihood of the exact sequence

$$L(p) = p^k(1-p)^{n-k}$$

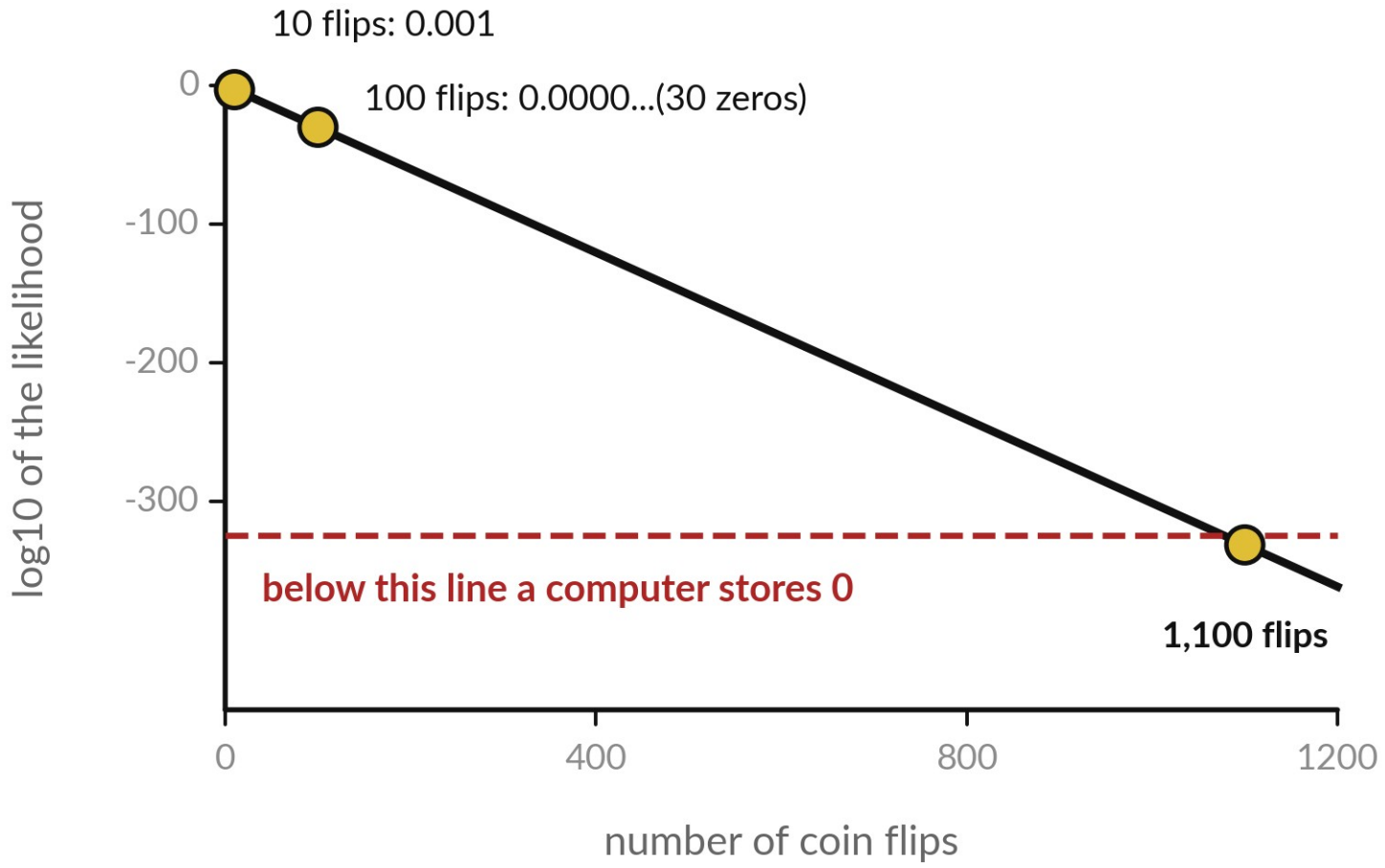
Maximum likelihood estimate

$$\hat{p} = \frac{k}{n} = \frac{7}{10} = 0.7$$

Symbol	What it means	In our data
p	The chance of heads. This is the unknown parameter.	tried from 0.1 to 0.9
k	The number of heads.	7
n	The number of flips.	10
p-hat	The maximum likelihood estimate of p.	0.7

Why: the slope of ln L, k/p - (n - k)/(1 - p), is zero at p = k/n. Counting the orders (a factor 120) does not move the peak.

A product of many small probabilities vanishes, so we add logarithms instead

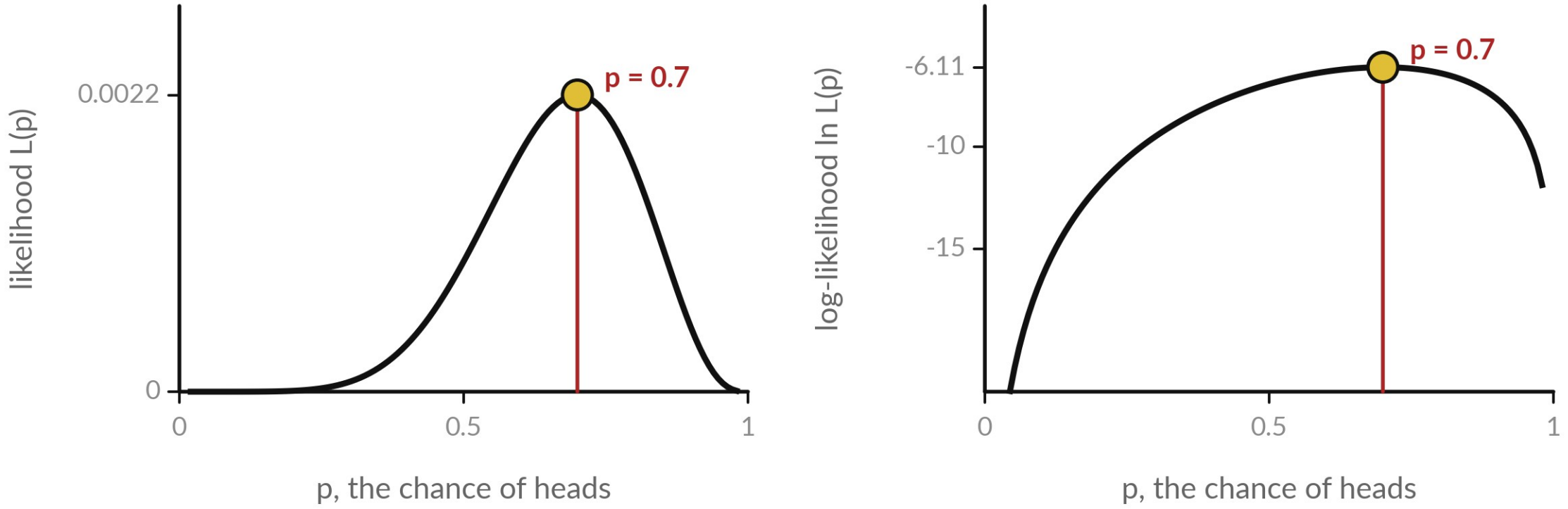


The log-likelihood is the natural logarithm (ln) of the likelihood.

A logarithm turns a product into a sum, and the maximum stays at the same p.

0.5 to the power 1,100 is stored as 0.0 by a computer. But 1,100 × ln(0.5) = -762.46 is a normal number.

The log-likelihood curve peaks at the same p as the likelihood curve



Left: likelihood. Right: log-likelihood. Same maximum, p = 0.7.

The log-likelihood is a sum, and training minimises its negative

Log-likelihood

$$\ln L(\theta) = \sum_{i=1}^{n} \ln P(y_i \mid \theta)$$

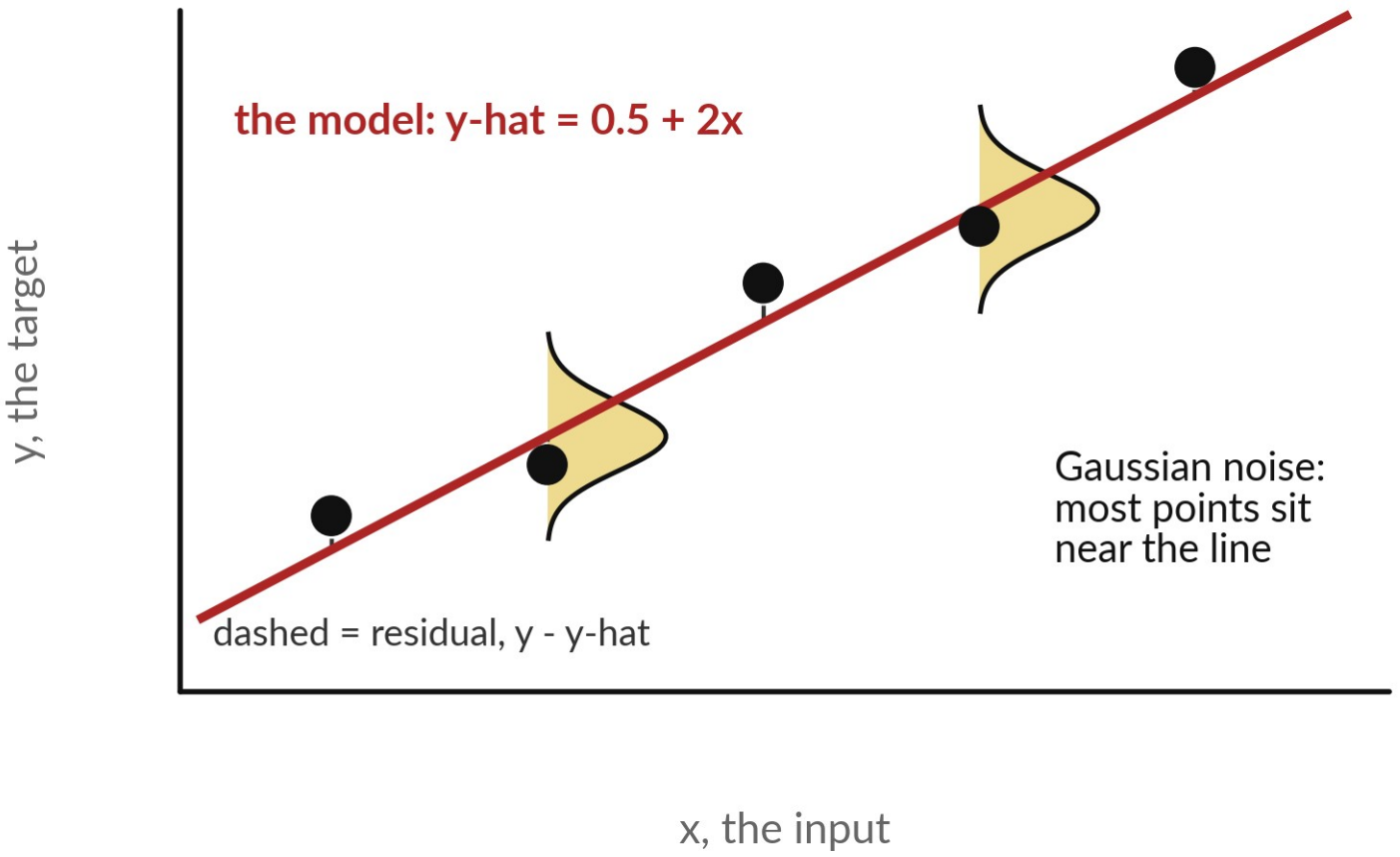
Loss (negative log-likelihood)

$$\mathrm{NLL}(\theta) = -\sum_{i=1}^{n} \ln P(y_i \mid \theta)$$

Symbol	What it means	In our data
theta	The parameters of the model.	p for the coin, mu for the Gaussian
y_i	The i-th data point.	one flip or one measurement
P(y_i | theta)	The probability the model gives to y_i.	p or 1 - p for a flip
n	The number of data points.	10 flips, 8 measurements

Minimising the NLL gives the same theta as maximising the likelihood. The letter p stays the coin.

Gaussian noise: each target sits near the prediction, with bell-shaped scatter



Assume y = y-hat + noise, where the noise follows a Gaussian (normal) distribution with mean 0 and standard deviation sigma.

Small residuals (y minus y-hat) are likely. Large ones are rare.

The yellow bells are the Gaussian probability density function drawn around the line.

Gaussian noise turns the negative log-likelihood into a squared error plus a constant

Gaussian density

$$p(y \mid \hat{y}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-\hat{y})^2}{2\sigma^2}\right)$$

Negative log-likelihood

$$\mathrm{NLL} = \frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \frac{n}{2} \ln(2\pi\sigma^2)$$

Symbol	What it means	In our data
y_i - y-hat_i	The residual of row i.	actual minus predicted
sigma	The standard deviation of the noise.	1 mm in the lab
Sum of squares	The sum of squared errors (SSE).	5.34 at mu = 10.25
Last term	A constant that does not depend on the model.	7.35 for n = 8, sigma = 1

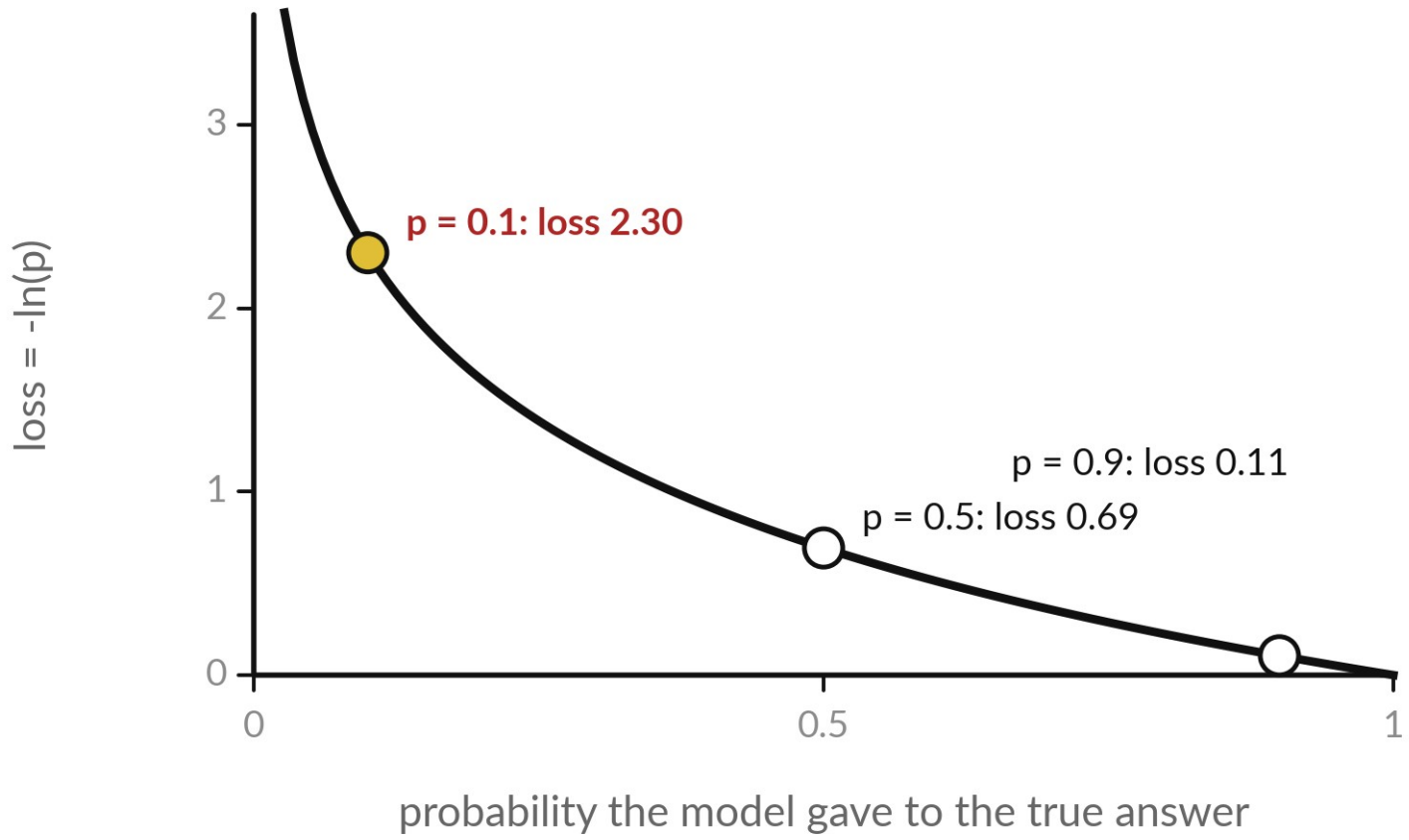
Only the squared-error part depends on the model. With the same sigma for every row, squared error is Gaussian MLE.

Eight measurements: the mu with the smallest squared error has the largest likelihood

mu (mm)	Sum of squared errors	Log-likelihood
9.00	17.84	-16.2715
9.50	9.84	-12.2715
10.00	5.84	-10.2715
10.25	5.34	-10.0215
10.50	5.84	-10.2715
11.00	9.84	-12.2715
11.50	17.84	-16.2715

Eight widths (mm, illustrative): 9.8, 10.4, 11.2, 9.1, 10.7, 10.1, 11.5, 9.2. Sigma = 1. The sample mean is 10.25. In every row, -(log-likelihood) = SSE / 2 + 7.35.

Cross-entropy: the loss -ln(p) grows steeply when the true answer gets a small probability



A Bernoulli model gives probability p to class 1 and 1 - p to class 0.

The negative log-likelihood of one label is the binary cross-entropy loss.

With many classes, the loss is -ln of the probability given to the true class.

A Bernoulli or categorical assumption gives the cross-entropy loss

Binary cross-entropy $$-[\, y \ln p + (1 - y) \ln(1 - p) \,]$$

Categorical cross-entropy $$-\ln p_c$$

Symbol	What it means	In our data
y	The true label, 0 or 1.	four rows: 1, 0, 1, 1
p	The probability the model gives to class 1.	four rows: 0.8, 0.3, 0.6, 0.9
c	The index of the true class.	the first of three classes
p_c	The probability given to the true class.	0.7

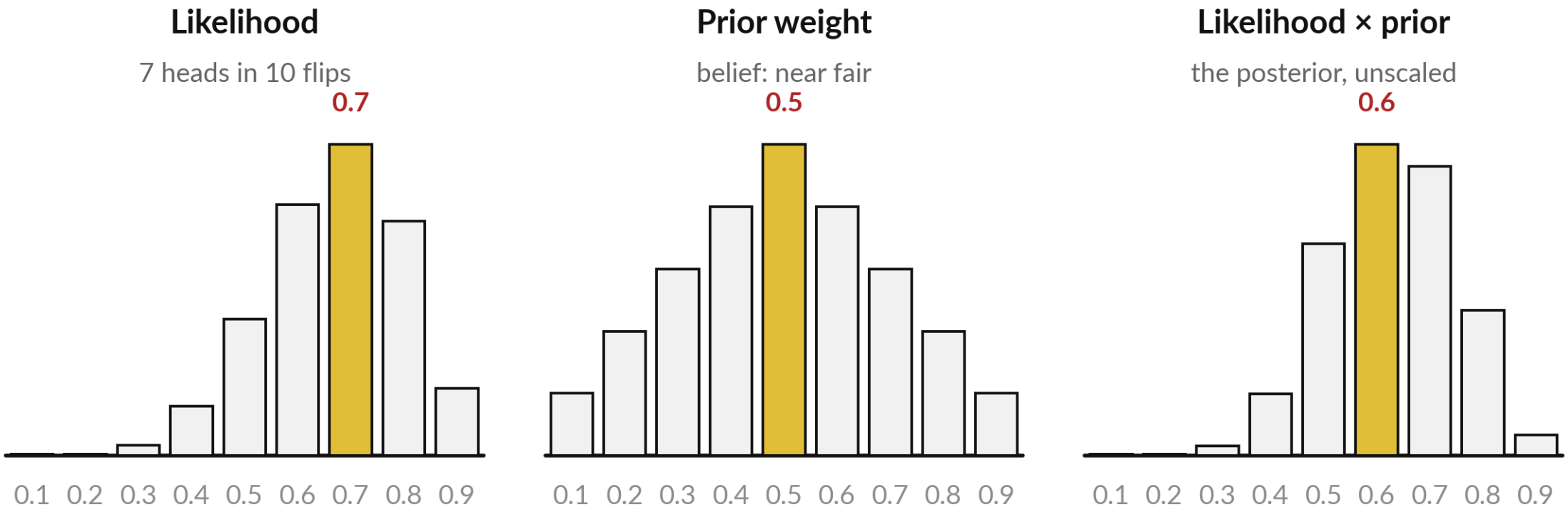
Illustrative. The four rows give a mean loss of 0.299. Classes 0.7, 0.2, 0.1 with the first true: 0.357.

Each common loss is a probability assumption in disguise

	Task	Assumption about the target	Loss that follows
	**Regression**	Gaussian noise around the prediction, the same sigma for every row	Squared error (mean squared error)
	**Yes or no**	Bernoulli: yes with probability p	Binary cross-entropy
	**One of K classes, such as the next token**	Categorical: K probabilities that add to 1	Categorical cross-entropy

Change the assumption and the right loss changes with it. For noise with heavy tails the log-likelihood gives a different loss.

MAP: the posterior is the likelihood times a prior, and the prior pulls the estimate



Gold = largest bar. Likelihood alone: p = 0.7 (MLE). With the prior: p = 0.6 (MAP).

By hand: likelihood times prior weight is largest at p = 0.6, so MAP is 0.6 and MLE is 0.7

p	Likelihood	Prior weight	Product
0.1	0.000000	1	0.000000
0.2	0.000007	2	0.000013
0.3	0.000075	3	0.000225
0.4	0.000354	4	0.001416
0.5	0.000977	5	0.004883

p	Likelihood	Prior weight	Product
0.6	0.001792	4	0.007166
0.7	0.002224	3	0.006671
0.8	0.001678	2	0.003355
0.9	0.000478	1	0.000478

A Gaussian prior on the weights adds an L2 penalty, the idea behind weight decay

MAP loss

$$-\ln p(\text{data} \mid w) - \ln p(w) = \frac{1}{2\sigma^2}\text{SSE} + \frac{1}{2\tau^2}\sum_j w_j^2 + \text{constant}$$

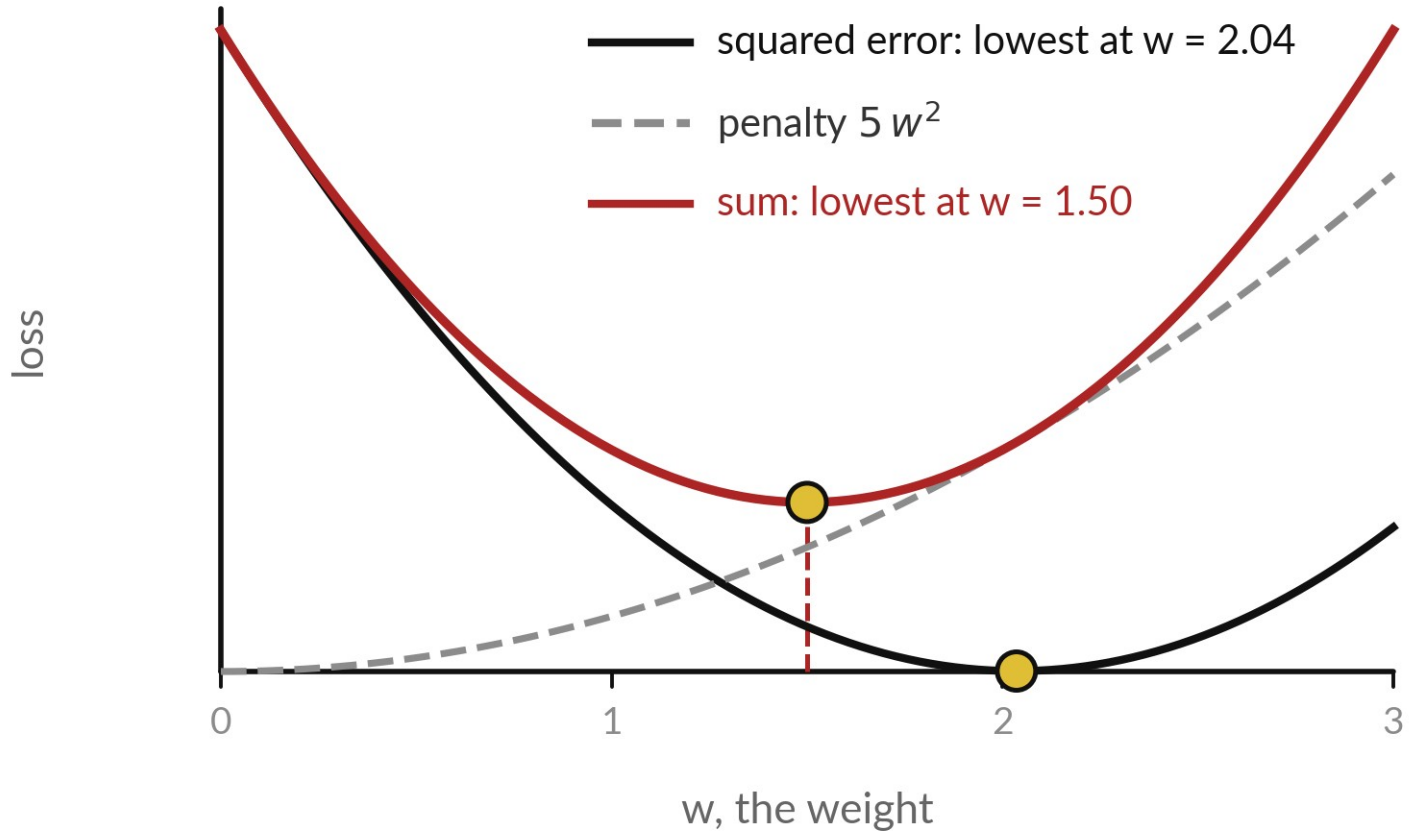
Multiply by 2 sigma^2: same minimum as

$$\text{SSE} + \lambda \sum_j w_j^2, \qquad \lambda = \frac{\sigma^2}{\tau^2}$$

Symbol	What it means	In our data
w_j	The j-th weight of the model.	one weight in our example
tau	Standard deviation of the Gaussian prior on each weight. Small tau means a strong belief that w is near 0.	prior mean 0
lambda	The weight decay strength (L2 regularisation strength).	sigma squared over tau squared
SSE	The sum of squared errors.	from the Gaussian noise slide

Small tau, large lambda, smaller weights. Weight decay: exact for plain gradient descent.

By hand: the penalty pulls the best weight towards 0

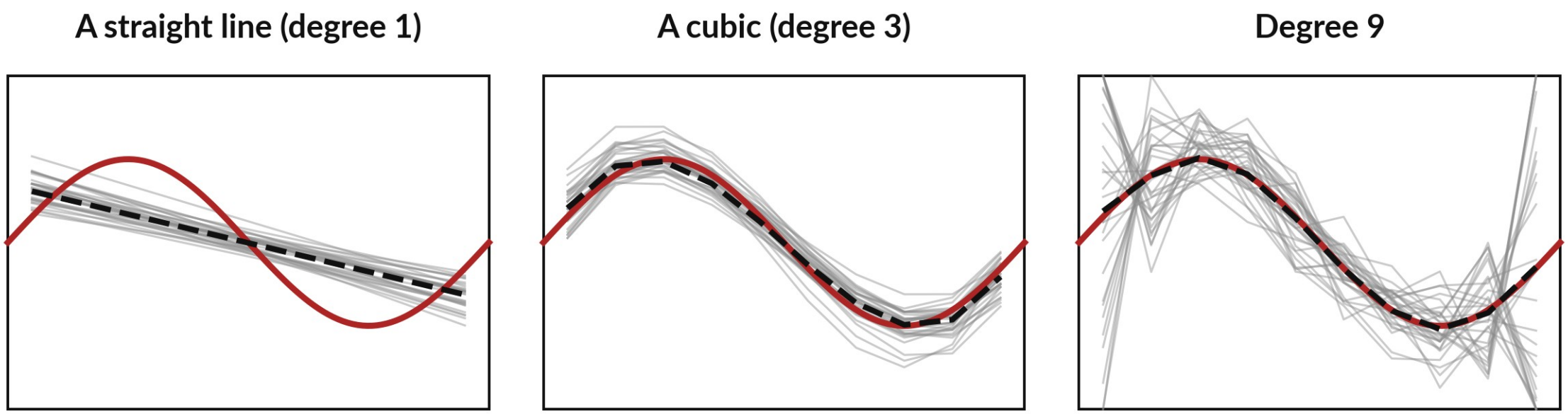


Fit y = w x on x = 1, 2, 3 and y = 2.1, 3.9, 6.2 (illustrative). The best w is 28.5 / (14 + lambda).

lambda = 0, 1, 5, 14 gives w = 2.036, 1.9, 1.5, 1.018.

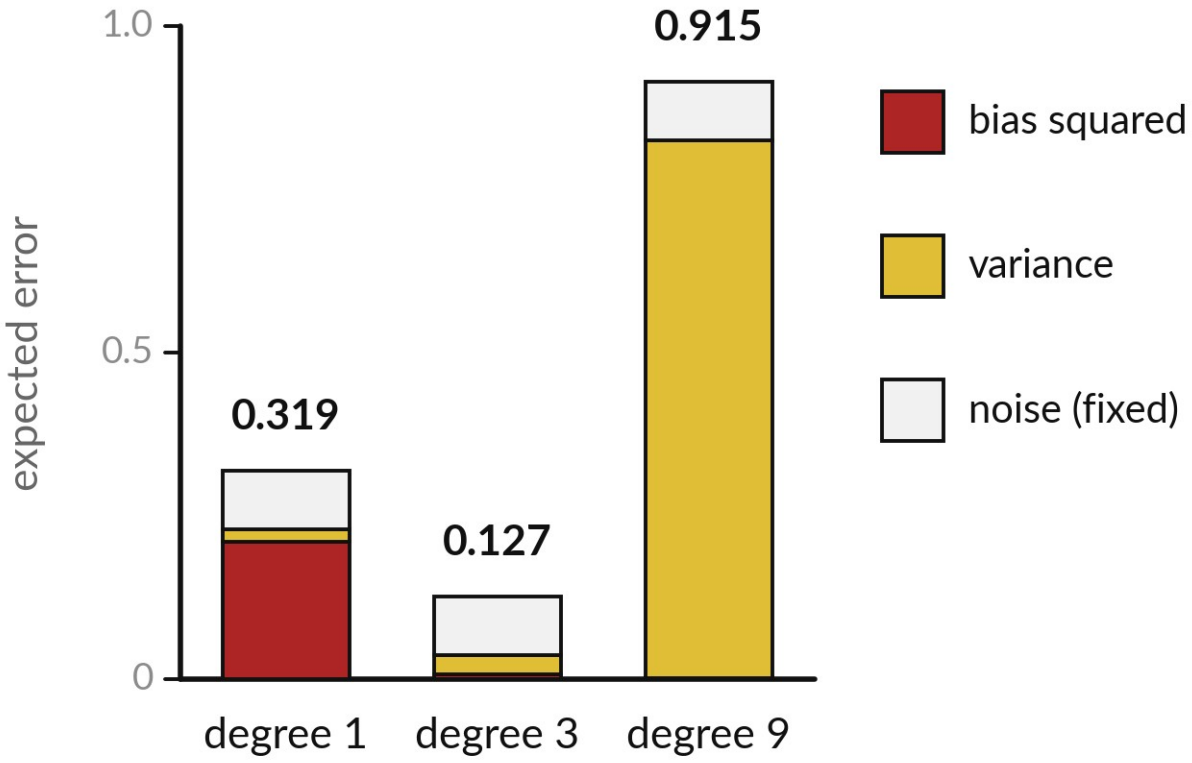
The sum of the two curves has its lowest point to the left of the squared-error minimum.

Bias and variance: refit the same model on 300 fresh noisy training sets



Truth y = sin(2 pi x) plus noise 0.3, 10 points per set. Left: high bias. Right: high variance.

Expected error splits into bias squared, variance and noise



The cubic has the smallest expected error, 0.127. It balances bias and variance.

Degree 1: bias squared 0.211, variance 0.018.

Degree 9: bias squared 0.001, variance 0.824.

The noise, 0.3 squared = 0.09, stays in every bar.

The trade-off as a formula: expected squared error = bias squared + variance + noise

Decomposition

$$\text{Error} = \text{Bias}^2 + \text{Variance} + \sigma^2$$

Cubic, from the simulation

$$0.127 = 0.007 + 0.030 + 0.09$$

Symbol	What it means	In our data
Bias squared	The squared gap between the average prediction and the truth.	degree 1: 0.211
Variance	How much the prediction moves between training sets.	degree 9: 0.824
sigma squared	The variance of the noise. It cannot be removed.	0.3 squared = 0.09
Error	The expected squared error on new data.	cubic: 0.127

A larger lambda lowers variance and raises bias. Choose lambda on held-out data.

Three common mistakes about likelihood and MAP

The mistake

Likelihood of p = 0.7 means the probability that p is 0.7.

3 heads in 3 flips proves p = 1.

Squared error assumes nothing.

The fix

It is how probable the data are if p is 0.7.

Little data makes MLE extreme. A prior tempers it.

It assumes Gaussian noise.

Every loss hides a probability model. Name it.

Read the recipe before you write it: grid, likelihood, logarithm, argmax

```
k, n = 7, 10
prior = np.array([1, 2, 3, 4, 5, 4, 3, 2, 1])
p = np.linspace(0.1, 0.9, 9)
L = p**k * (1 - p)**(n - k)
logL = k*np.log(p) + (n - k)*np.log(1 - p)
p_hat = p[np.argmax(logL)]
p_map = p[np.argmax(L * prior)]
```

In plain words

Lines 1, 2: the data and the prior weights.

Line 3: nine candidate values of p.

Line 4: the likelihood for every p.

Line 5: the log-likelihood, a sum.

Line 6: argmax gives the position of the largest value: the MLE.

Line 7: times the prior: the MAP estimate.

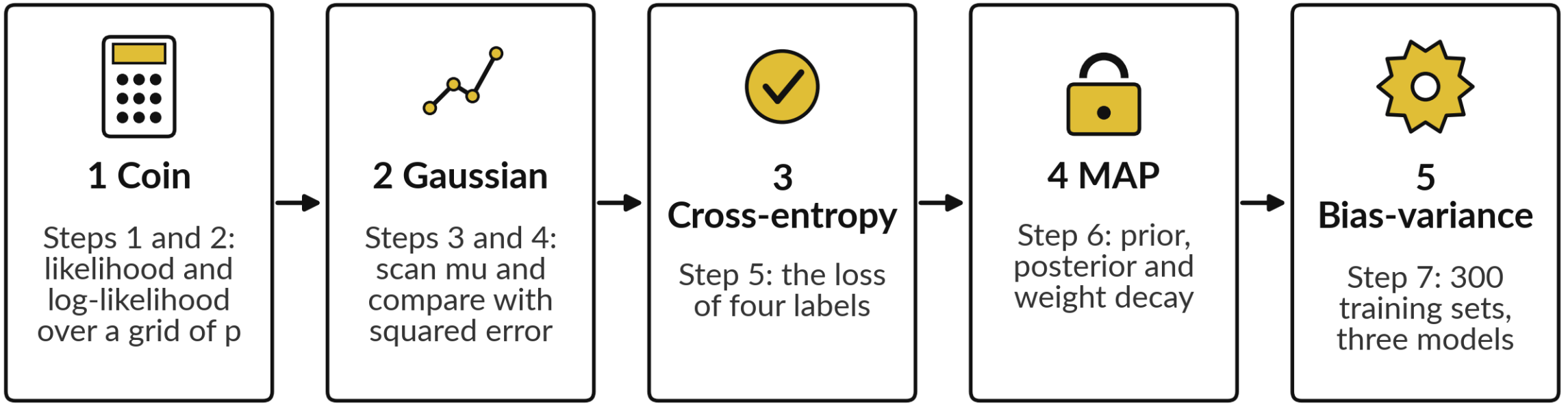
Spreadsheet: INDEX and MATCH on MAX return the p.



Part 2: the lab

Close the slides. Open the notebook or the workbook.

The lab has five parts: coin, Gaussian, cross-entropy, MAP and bias-variance



Python: Steps 1 to 7. Spreadsheet: sheets Coin, Gaussian, Shrink (cross-entropy and weight decay) and Bias-variance. Predict, run, then change one thing.

Check your numbers against these before you move on

	What you compute	You should see
	Likelihood of this exact sequence (7 heads, 3 tails) at p = 0.7	0.002224
	Probability of 7 heads in any order at p = 0.7	0.2668
	Log-likelihood of the exact sequence at p = 0.7	-6.1086
	Best mu for the eight widths, and its log-likelihood	10.25 and -10.0215
	Binary cross-entropy of the four labels (1, 0, 1, 1; p = 0.8, 0.3, 0.6, 0.9)	0.2990
	MAP estimate of p	0.6
	Weights for lambda = 0, 1, 5, 14	2.036, 1.9, 1.5, 1.018
	Expected error, degrees 1, 3 and 9 (Python)	0.319, 0.127, 0.915

If your number differs, stop and ask. Find the step that went wrong. Do not change a number to match.

Errors you will meet, what they mean and what to do

	What you see	What it means	What to do
	**NameError**	A name from a cell you have not run yet	Run all cells from the top, in order
	**TypeError: NoneType**	An answer cell (marked gold) still says None	Replace None with your code and run the cell again
	**likelihood prints 0.0 or nan**	A product underflowed, or log of 0	Work with the log-likelihood and keep p between 0 and 1
	**Spreadsheet shows #NAME?**	A function name is mistyped	Check the spelling, such as NORM.DIST

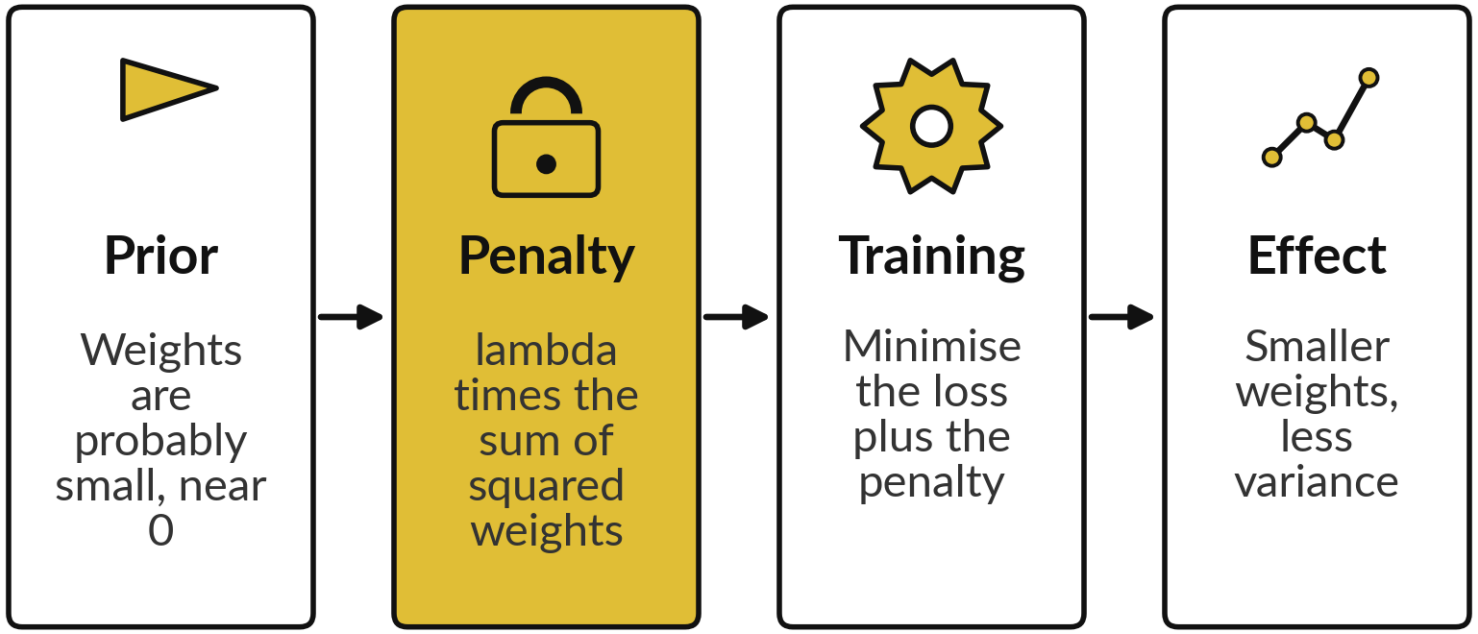
Where this appears in AI: the likelihood of the next token

the next word	The	cat	sat	down	total
probability the model gave it	0.2	0.1	0.25	0.05	
-ln(p)	1.61	2.30	1.39	3.00	8.29

Illustrative (made-up numbers to show the arithmetic): a 4-token sentence.

Published research: training minimises the sum of -ln p. Here the sum is 8.29.

Where this appears in AI: weight decay and the Gaussian prior



The Llama 1 and 2 papers (Touvron et al., 2023) report AdamW with weight decay 0.1.

With plain gradient descent, weight decay equals an L2 penalty (MAP, Gaussian prior). AdamW (Loshchilov and Hutter, 2019) decays separately: an analogy.

ChatGPT, Gemini and Claude do not publish theirs: no claim.

Check an AI assistant's answer against your own numbers

5 minutes Optional, with any assistant

Ask an assistant for the MLE of p after 7 heads in 10 flips, and the likelihood of that sequence.

Compare with 0.7 and 0.002224 (any order: 0.2668).

Ask for the MAP estimate on the grid p = 0.1 to 0.9 with the prior weights from today. Compare with 0.6.

Never paste confidential data into an assistant.

You keep

A note: where the assistant agreed with your numbers, and where it did not.

Exit check: three questions and their short answers

Question	Short answer
A coin gives 9 heads in 12 flips. What is the MLE of p?	k / n = 9 / 12 = 0.75.
Which probability assumption turns maximum likelihood into squared error?	Gaussian noise around the prediction.
What does a Gaussian prior on the weights add to the loss?	An L2 penalty, lambda times the sum of squared weights (weight decay).

Answer each in one sentence before you read the right-hand column.

Save your lab and keep the five numbers

What you keep from today

Keep your notebook or workbook with all checks passing. Keep the five numbers: MLE 0.7, MAP 0.6, best mu 10.25, cross-entropy 0.299 and the three expected errors of the bias-variance simulation.

Our question today

When a language model is trained, what exactly is it trying to make as large as possible?

Next session

Session 17: information theory. Entropy, cross-entropy, KL divergence, softmax and perplexity.

Take away

MLE and MAP
0.7 and 0.6

Best mu
10.25

Cross-entropy
0.299

Next session
Information theory